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CHAPTER 9
FLIP-FLOPS

9.1 Introduction

The digital circuits considered thus far have been combinational, i.e., the outputs at any instant of time are entirely dependent upon the inputs present at that time. Although every digital system is likely to have combinational circuits, most systems encountered in practice also include memory elements, which require that the system be described in terms of *sequential logic*.

A block diagram of a sequential circuit is shown in Figure 9.1. It consists of combinational circuits that accept external inputs and inputs from outputs of memory elements and generate external outputs and inputs to memory elements referred to as excitation.

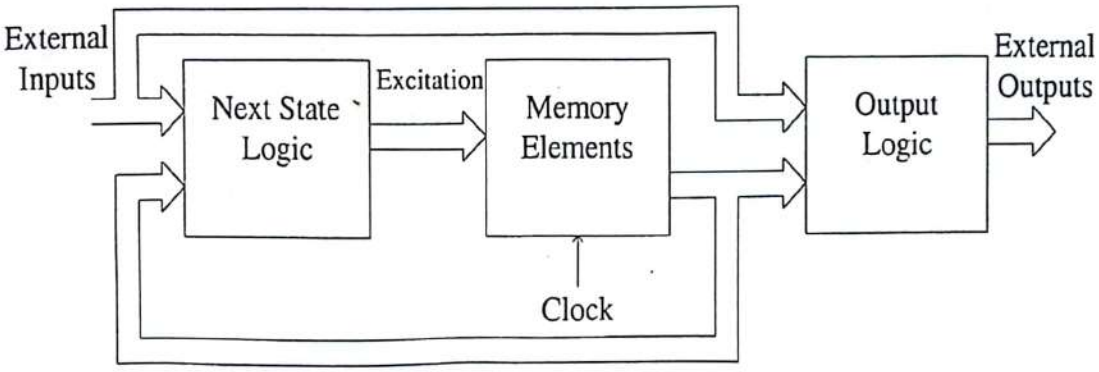


Figure 9.1 Block diagram of a sequential circuit.

A memory element is a device in which one bit of information (1 or 0) can be stored or retained until necessary, and thereafter its content can be replaced by a new value. The contents of memory elements in Figure 9.1 can be changed by the outputs of the combinational circuit which are connected to its input.

The combinational circuit performs certain operations, some of which are used to determine the value to be stored in memory elements. The other operations are performed on external inputs and memory outputs to generate the external outputs.

The above process demonstrates the dependence of the external outputs of a sequential circuit on the external inputs and the present contents of the memory elements (referred to as the *present state* of the sequential circuit). The new contents of the memory elements (referred to as the *next state* of the sequential circuit) depend on the external inputs and the present state. Hence, the output of a sequential circuit is a function of the time sequence of inputs and the *internal states*.

Sequential circuits are classified into two main categories, known as *asynchronous* and *synchronous* sequential circuits depending on timing of their signals.

A sequential circuit whose behavior depends upon the sequence in which the input signals change is referred to as an *asynchronous* sequential circuit. The outputs will be affected whenever the inputs change. The commonly used memory elements

in these circuits are time delay devices. These can be regarded as combinational circuits with feedback. Because of the feedback among logic gates, an asynchronous circuit may, at times, become unstable. The instability problem imposes many difficulties on the designer. Hence they are not as commonly used as synchronous sequential circuits.

A sequential circuit whose behavior can be defined from the knowledge of its signal at discrete instants of time is referred to as a *synchronous* sequential circuit. In these circuits, the memory elements are affected only at discrete instants of time. The synchronization is achieved by a timing device known as a *system clock* which generates a periodic train of clock pulses as shown in Figure 9.2. The outputs are affected only with the application of a clock pulse. Synchronous circuits have gained considerable domination and wide popularity and are also known as *clocked-sequential circuits*. They do not manifest instability problems. The memory elements used in clocked-sequential circuits are flip-flops that are capable of storing binary information.

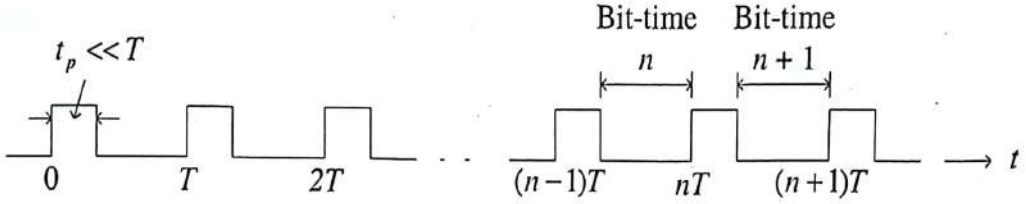


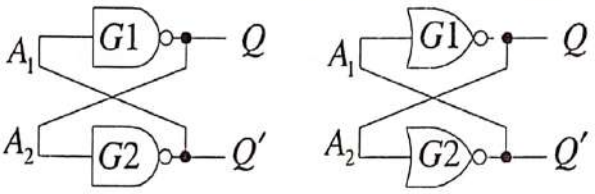
Figure 9.2 A train of pulses.

In this chapter we examine the various types of flip-flops and define their logical properties.

9.2 A 1-Bit Memory Cell

The basic digital memory circuit is known as flip-flop. It has two stable states that are known as the 1 state and the 0 state. It can be obtained by using NAND or NOR gates.

We will be systematically developing a flip-flop circuit starting from the fundamental circuit shown in Figure 9.3. It consists of two inverters $G1$ and $G2$ (NAND and NOR gates used as inverters). The output of $G1$ is connected to the input of $G2$ (A_2) and the output of $G2$ is connected to the input of $G1$ (A_1).



(a) Using NAND gates (b) Using NOR gates

Figure 9.3 Cross-coupled inverters as a memory element.

Let us assume the output of $G1$ to be $Q = 1$, which is also the input of $G2$ ($A_2 = 1$). Therefore, the output of $G2$ will be $Q' = 0$, which makes $A_1 = 0$ and

consequently $Q = 1$ which confirms our assumption. In a similar manner, it can be demonstrated that if $Q = 0$, then $Q' = 1$ and this is also consistent with the circuit connections. From the above discussion we note the following:

1. The outputs Q and Q' are always complementary.
2. The circuit has two stable states; in one of the stable state $Q = 1$, which is referred to as the 1 state (or *set state*), whereas in the other stable state $Q = 0$, which is referred to as the 0 state (or *reset state*).
3. If the circuit is in 1 state, it continues to remain in this state and similarly if it is in 0 state, it continues to remain in this state. This property of the circuit is referred to as memory, i.e., it can store 1-bit of digital information.

Since this information is locked or latched in this circuit, this circuit is also referred to as a *latch*.

S-R Latch

In the latch of Figure 9.3, there is no way of entering the desired digital information to be stored in it. In fact, when the power is switched on, the circuit switches to one of the stable states ($Q = 1$ or $Q = 0$) and it is not possible to predict the state. If we replace the inverters *G*1 and *G*2 with 2-input NAND gates as in Figure 9.4(a) and with 2-input NOR gates as in Figure 9.4(b), the other input terminals of the gates can be used to enter the desired digital information. Two additional inverters *G*3 and *G*4 have been added in Figure 9.4(a) for reasons that will become clear from the following discussion.

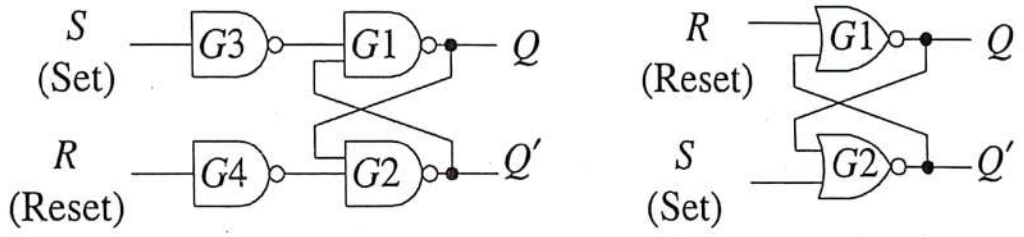


Figure 9.4 The memory cell with provision for entering data.

If $S = R = 0$, the circuit are exactly the same as that of Figure 9.3. If $S = 1$ and $R = 0$, in Figure 9.4(a), the outputs of *G*3 will be 0 and the output of *G*4 will be 1. Since one of the inputs of the NAND gate *G*1 is 0, its output will certainly be 1. Consequently, both the inputs of the NAND gate *G*2 will be 1 giving an output $Q' = 0$. In Figure 9.4(b), since one of the inputs of the NOR gate *G*2 is 1, its output will certainly be 0. Consequently, both the inputs of the NOR gate *G*1 will be 0 giving an output $Q = 1$. Hence, for this input condition, $Q = 1$ and $Q' = 0$.

Similarly, if $S = 0$ and $R = 1$ then outputs will be $Q = 0$ and $Q' = 1$. The first of these two input conditions ($S = 1$ and $R = 0$) makes $Q = 1$, which is referred to as the *set state*, whereas the second input condition ($S = 0$ and $R = 1$) makes $Q = 0$, which is referred to as the *reset state* or *clear state*. This gives us the means for entering the desired bit in the latch.

Now we see what happens if the input conditions are changed from $S = 1$ and $R = 0$ to $S = R = 0$ or from $S = 0$ and $R = 1$ to $S = R = 0$. The output remains unaltered. This shows the basic difference between a combinational circuit and a sequential circuit, even though the sequential circuit is made up of combinational circuits.

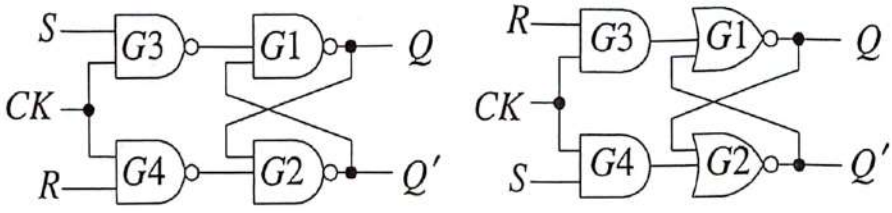
The two input terminals are designated as set (S) and reset (R) because $S = 1$ brings the circuit in set state and $R = 1$ brings it to reset or clear state.

If $S = R = 1$, in Figure 9.4(a) both the outputs Q and Q' will try to become 1 and in Figure 9.4(b) both the outputs Q and Q' will try to become 0, which is not allowed and, therefore, this input condition is prohibited.

9.3 S-R Flip-Flop

9.3.1 Basic S-R Flip-Flop

It is often required to set or reset the memory cell of Figure 9.4 in synchronization with a train of pulses (Figure 9.2) known as clock (abbreviated as *CK*). Such circuit is shown in Figure 9.5 and is referred to as a clocked *set-reset (S-R) flip-flop*.



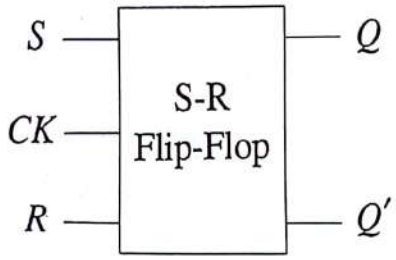
(a) Using NAND latch (b) Using NOR latch

Figure 9.5 Logic diagram of S-R Flip-Flop.

In this circuit, if a clock pulse is present ($CK = 1$), its operation is exactly the same as that of Figure 9.4. On the other hand, when the clock pulse is not present ($CK = 0$), in Figure 9.5(a), the outputs of gates *G*3 and *G*4 are 1 irrespective of the values of *S* or *R* and the outputs of gates *G*1 and *G*2 are unchanged. When $CK = 0$, in Figure 9.5(b), the outputs of gates *G*3 and *G*4 are 0 irrespective of the values of *S* or *R* and the outputs of gates *G*1 and *G*2 are unchanged. In general, the circuits of Figure 9.5 respond to the inputs *S* and *R* only when the clock is present.

The logic symbol of S-R flip-flop is shown in Figure 9.6(a). Assuming that the inputs do not change during the presence of the clock pulse, we can express the operation of the S-R flip-flop as shown in the table of Figure 9.6(b), which is known

as characteristic table (also known as truth table). Here S_n and R_n denote the inputs and Q_n the output during the bit time n (Figure 9.2). Q_{n+1} denotes the output after the pulse passes, i.e., in the bit time $n+1$. Figure 9.6(c) shows the reduced characteristic table of S-R flip-flop derived from Figure 9.6(b).



(a) Logic symbol

S_n	R_n	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	?
1	1	1	?

(b) Characteristic table

S_n	R_n	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	?

(c) Reduced characteristic table

Figure 9.6 Logic symbol and characteristic table of S-R flip-flop.

If $S_n = R_n = 0$, and the clock pulse is applied, the output at the end of the clock pulse is same as the output before the clock pulse, i.e., $Q_{n+1} = Q_n$. This is indicated in the first and second rows of Figure 9.6(b) and in the first row of Figure 9.6(c).

If $S_n = 0$ and $R_n = 1$, the output at the end of the clock pulse will be 0, i.e., $Q_{n+1} = 0$. This is indicated in the third and fourth rows of Figure 9.6(b) and in the second row of Figure 9.6(c).

If $S_n = 1$ and $R_n = 0$, the output at the end of the clock pulse will be 1, i.e., $Q_{n+1} = 1$. This is indicated in the fifth and sixth rows of Figure 9.6(b) and in the third row of Figure 9.6(c).

In the circuit of Figure 9.4, it was mentioned that $S = R = 1$ is prohibited. Let us see what happens in the S-R flip-flop of Figure 9.5 if $S_n = R_n = 1$. In Figure 9.5(a), when the clock is present the outputs of gates *G*3 and *G*4 are both 0, making one of the inputs of *G*1 and *G*2 NAND gates 0. Consequently, Q and Q' both will be 1, which is inconsistent with our assumption of complementary outputs. Now, when the clock pulse has passed away ($CK = 0$), the outputs of *G*3 and *G*4 will rise from 0 to 1. Depending upon the propagation delays of the gates, either the stable

states $Q_{n+1} = 1$ ($Q'_{n+1} = 0$) or $Q_{n+1} = 0$ ($Q'_{n+1} = 1$) will result. In Figure 9.5(b), when the clock is present the outputs of gates *G*3 and *G*4 are both 1, making one of the inputs of *G*1 and *G*2 NOR gates 1. Consequently, Q and Q' both will be 0, which is also inconsistent with the assumption of complementary outputs. Now, when the clock pulse has passed away ($CK = 0$), the outputs of *G*3 and *G*4 will go from 1 to 0. Depending upon the propagation delays of the gates, either the stable states $Q_{n+1} = 1$ ($Q'_{n+1} = 0$) or $Q_{n+1} = 0$ ($Q'_{n+1} = 1$) will result. That means, in both the Figure 9.5(a) and (b), when $S_n = R_n = 1$ the state of the circuit is undefined, indeterminate or ambiguous and therefore is indicated by a question mark in the seventh and eighth rows of Figure 9.6(b) and in the fourth row of Figure 9.6(c). Thus, the condition $S_n = R_n = 1$ is forbidden and it must not be allowed to occur.

The characteristic table of Figure 9.6(b) is shown in the K-map of Figure 9.7, which gives

$$Q_{n+1} = S_n + R'_n Q_n$$

$$S_n R_n = 0$$

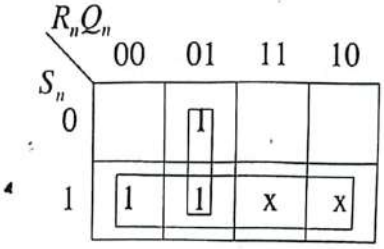


Figure 9.7 Characteristic equation of S-R flip-flop.

The first equation is known as the characteristic equation of S-R flip-flop. This equation specifies the value of the next state as a function of the present state and the inputs. The characteristic equation is an algebraic expression for the binary information of the characteristic table. The two indeterminate states are marked by x's in the K-map, since they may result in either a 1 or a 0. However, the relation $S_n R_n = 0$ must be included as part of the characteristic equation to specify that both S_n and R_n cannot equal 1 simultaneously.

In the design of sequential circuits, we usually come across situations in which the *present state* and the *next state* of the circuit are specified, and we have to find the input conditions that must prevail to cause the desired transition of the state. By the present state and the next state we mean the state of the circuit prior to and after the clock pulse respectively. For example, the output of an S-R flip-flop before the clock pulse $Q_n = 0$ and it is desired that the output does not change when the clock pulse is applied. What input conditions (S_n and R_n values) must exist to achieve this? From the characteristic table of Figure 9.6(b) we obtain the following conditions:

1. $S_n = R_n = 0$ (first row)
2. $S_n = 0$ and $R_n = 1$ (third row)

We conclude from the above conditions that the S_n input must be 0, whereas the R_n input may be either 0 or 1 (don't care). For finding input conditions for all possible situations, the characteristic table of Figure 9.6(b) is rearranged as in Table 9.1, from which the excitation table for S-R flip-flop of Table 9.2 is derived.

Table 9.1 Rearranged characteristic table of S-R flip-flop.

S_n	R_n	Q_n	Q_{n+1}
0	0	0	0
0	1	0	0
1	0	0	1
0	1	1	0
0	0	1	1
1	0	1	1
1	1	0	?
1	1	1	?

Table 9.2 Excitation table of S-R flip-flop.

Q_n	Q_{n+1}	S_n	R_n
0	0	0	x
0	1	1	0
1	0	0	1
1	1	x	0

9.3.2 S-R Flip-Flop with Preset and Clear Inputs

In the S-R flip-flop of Figure 9.5, when the power is switched on, the state of the circuit is uncertain. It may come to set ($Q = 1$) or reset ($Q = 0$) state. In many applications it is desired to initially set or reset the flip-flop, i.e., the initial sate of the flip-flop is to be assigned. This is accomplished by using the direct, or *asynchronous inputs*, referred to as *preset* (*Pr*) and *clear* (*Cr*) inputs. These inputs may be applied at any time between clock pulses and are not in synchronism with the clock. S-R flip-flop with preset and clear inputs is shown in Figure 9.8.

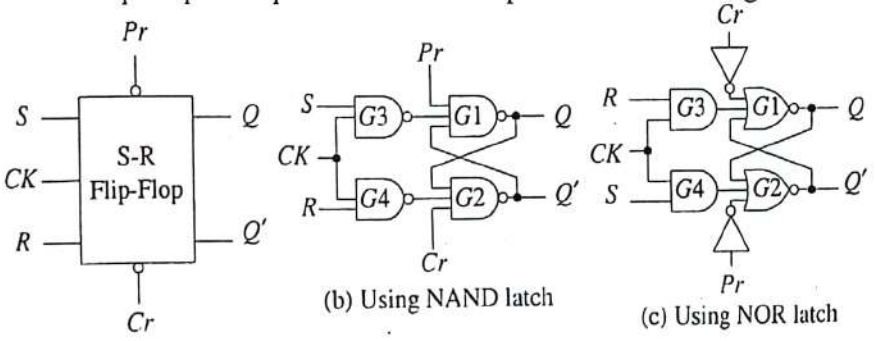


Figure 9.8 S-R flip-flop with preset and clear inputs.

If $Pr = Cr = 1$, the circuits operate in accordance with the characteristic table of S-R flip-flop given in Figure 9.6(b). If $Pr = 0$ and $Cr = 1$, in Figure 9.8(b), the output of $G1$ (Q) will certainly be 1. Consequently, all the three inputs to $G2$ will be 1, which will make $Q' = 0$. In Figure 9.8(c), the output of $G2$ (Q') will certainly be 0. Consequently, all the three inputs to $G1$ will be 0, which will make $Q = 1$. Hence, making $Pr = 0$ sets the S-R flip-flop.

Similarly, if $Pr = 1$ and $Cr = 0$, in Figure 9.8(b), the output of $G2$ (Q') will certainly be 1. Consequently, all the three inputs to $G1$ will be 1, which will make $Q = 0$. In Figure 9.8(c), the output of $G1$ (Q) will certainly be 0. Consequently, all the three inputs to $G2$ will be 0, which will make $Q' = 1$. Hence, making $Cr = 0$ resets the S-R flip-flop.

Once the state of the flip-flop is established asynchronously, the asynchronous inputs Pr and Cr must be connected to 1 before the next clock is applied. The condition $Pr = Cr = 0$ must not be used, since this leads to an uncertain state.

The logic symbol of S-R flip-flop with preset and clear inputs is shown in Figure 9.8(a). In the logic symbol bubbles are used for Pr and Cr inputs, which means these are active-LOW, i.e., the intended function is performed when the signal applied to Pr or Cr is LOW. The operation of S-R flip-flop is summarized in Table 9.3.

Table 9.3 Summary of operation of S-R flip-flop.

Inputs *CK* *Cr* *Pr*	Output *Q*	Operation performed
1 1 1	Q_{n+1} (Figure 9.6(c))	Normal flip-flop
0 0 1	0	Clear
0 1 0	1	Preset

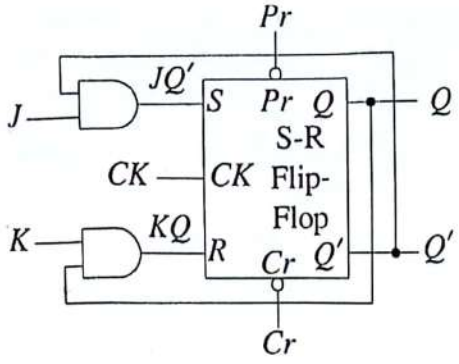
9.4 J-K Flip-Flop

9.4.1 Basic J-K Flip-Flop

The uncertainty in the state of S-R flip-flop when $S_n = R_n = 1$ (seventh and eighth rows of Figure 9.6(b)) can be eliminated by converting it into a J-K flip-flop. The data inputs are J and K, which are ANDed with Q' and Q, respectively, to obtain S and R inputs, i.e.,

$$S = J \cdot Q'$$
$$R = K \cdot Q$$

A J-K flip-flop thus obtained is shown in Figure 9.9(a) and its truth table is given in Figure 9.9(b).



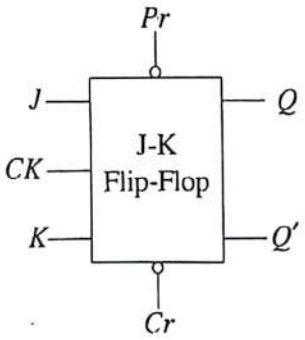
(a) Circuit connection

J_n	K_n	Q_n	S_n	R_n	Q_{n+1}
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	0	0
0	1	1	0	1	0
1	0	0	1	0	1
1	0	1	0	0	1
1	1	0	1	0	1
1	1	1	0	1	0

(b) Truth table

Figure 9.9 An S-R flip-flop converted into J-K flip-flop.

The logic symbol of a J-K flip-flop is shown in Figure 9.10(a). The characteristic table of a J-K flip-flop is shown in Figure 9.10(b) and a reduced characteristic table is also shown in Figure 9.10(c) for convenience.



(a) Logic symbol

J_n	K_n	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

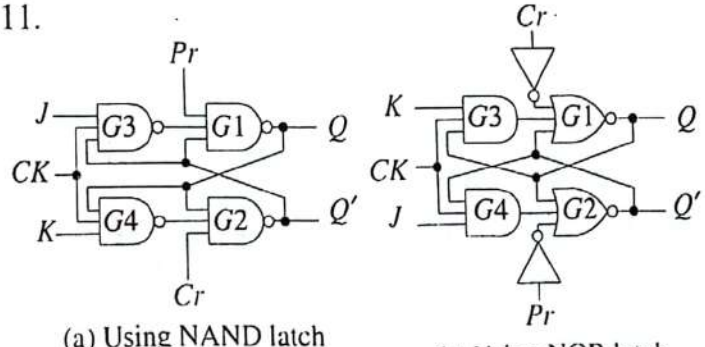
(b) Characteristic table

J_n	K_n	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	Q'_n

(c) Reduced characteristic table

Figure 9.10 Logic symbol and characteristic table of J-K flip-flop.

It is not necessary to use the AND gates of Figure 9.9(a), since the same function can be performed by adding an extra input terminal to each NAND gate *G*3 and *G*4 of Figure 9.8(b) or to each AND gate *G*3 and *G*4 of Figure 9.8(c). With this modification incorporated in Figure 9.8(b) and (c), we obtain the J-K flip-flop as shown in Figure 9.11.



(a) Using NAND latch

(b) Using NOR latch

Figure 9.11 J-K flip-flop.

The characteristic table of Figure 9.10(b) is shown in the K-map of Figure 9.12, which gives the following characteristic equation of J-K flip-flop.

$$Q_{n+1} = J_n Q_n' + K_n' Q_n$$

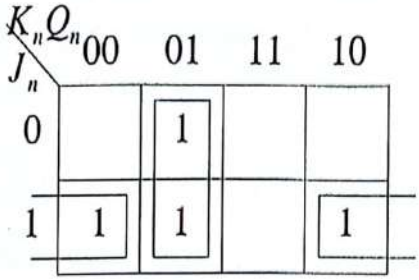


Figure 9.12 Characteristic equation of J-K flip-flop.

The characteristic table of Figure 9.10(b) is rearranged in Table 9.4 for obtaining the excitation table of J-K flip-flop. The excitation table of J-K flip-flop derived from Table 9.4 is given in Table 9.5.

Table 9.4 Rearranged characteristic table of J-K flip-flop for obtaining excitation table.

J_nK_n	Q_nQ_{n+1}
0 0	0 0
0 1	0 0
1 0	0 1
1 1	0 1
0 1	1 0
1 1	1 0
0 0	1 1
1 0	1 1

Table 9.5 Excitation table of J-K flip-flop.

Q_nQ_{n+1}	J_nK_n
0 0	0 x
0 1	1 x
1 0	x 1
1 1	x 0

9.4.2 The Race-Around Condition in J-K Flip-Flop

The difficulty of both inputs 1 ($S = R = 1$) being not allowed in an S-R flip-flop is eliminated in a J-K flip-flop by using the feed-back connection from outputs to the inputs of the gates $G3$ and $G4$ in Figure 9.11. Characteristic table of Figure 9.10(b) assumes that the inputs do not change during the clock pulse ($CK = 1$), which is

not true because of the feedback connections. Consider, for example, that the inputs are $J = K = 1$ and $Q = 0$ and a pulse as shown in Figure 9.13 is applied at the clock input. After a time interval Δt equal to the propagation delay through two gates in series (NAND-NAND gates in Figure 9.11(a) and AND-NOR gates in Figure 9.11(b)), the output will change to $Q = 1$ (see seventh row of Figure 9.10(b)). Now we have $J = K = 1$ and $Q = 1$ and after another time interval of Δt the output will change back to $Q = 0$. Hence, we conclude that for the duration t_p of the clock pulse, the output will oscillate back and forth between 0 and 1. At the end of the clock pulse, the value of Q is uncertain. This situation is referred to as the *race-around condition*.

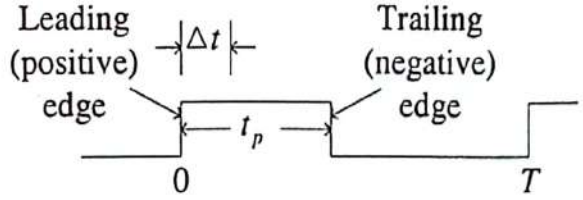


Figure 9.13 A clock pulse.

The race-around condition can be avoided if $t_p < \Delta t < T$. However, it may be difficult to satisfy this inequality because of very small propagation delays in ICs. A more practical method for overcoming this difficulty is the use of the master-slave (M-S) configuration discussed below.

9.4.3 The Master-Slave J-K Flip-Flop

A master-slave J-K flip-flop is a cascade of two S-R flip-flops with feedback from the outputs of the second to the inputs of the first as shown in Figure 9.14. Positive clock pulses are applied to the first flip-flop and the clock pulses are inverted before these are applied to the second flip-flop. (Master-slave J-K flip-flop circuit is presented here with NAND S-R flip-flop. The same circuit with NOR S-R flip-flop is left to the reader as problem.)

When $CK = 1$, the first flip-flop is enabled and the outputs Q_M and Q'_M respond to the inputs J and K according to the table of Figure 9.10(b). At this time, the second flip-flop is inhibited because its clock is LOW ($CK' = 0$). When CK goes LOW ($CK' = 0$), the first flip-flop is inhibited and the second flip-flop is enabled, because now its clock is HIGH ($CK' = 1$). Therefore, the outputs Q_S and Q'_S follow the outputs Q_M and Q'_M, respectively (second and third rows of Figure 9.6(c)). Since the second flip-flop simply follows the first one, it is referred to as the slave and the first one as the master. Hence, this configuration is referred to as *master-slave (M-S) flip-flop*.

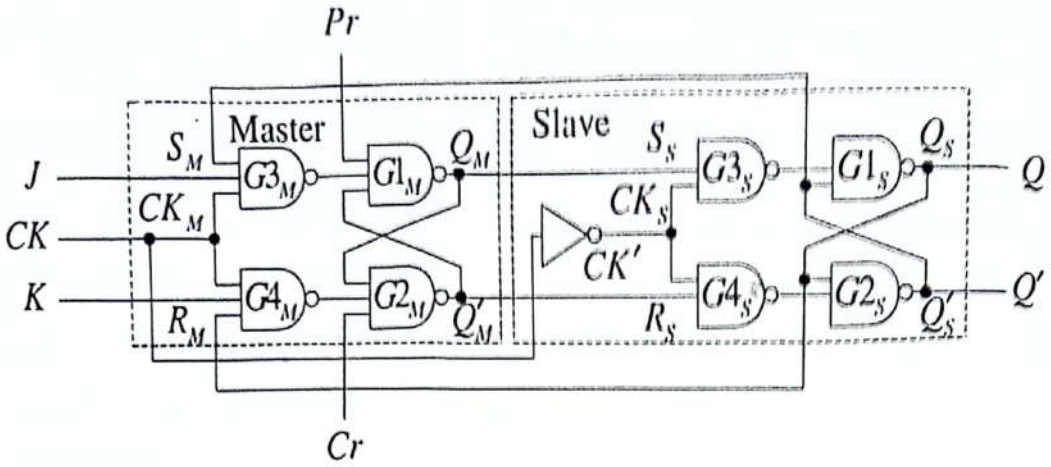


Figure 9.14 Master-slave J-K flip-flop using NAND S-R flip-flop.

In this circuit, the inputs to the gates $G3_M$ and $G4_M$ do not change during the clock pulse, therefore the race-around condition does not exist. The state of the master-slave flip-flop changes at the negative transition (trailing edge) of the clock pulse. The logic symbol of a M-S J-K flip-flop is shown in Figure 9.15. At the clock input terminal, the symbol > is used to illustrate that the output changes when the clock makes a transition and the accompanying bubble signifies negative transition (change in *CK* from 1 to 0).

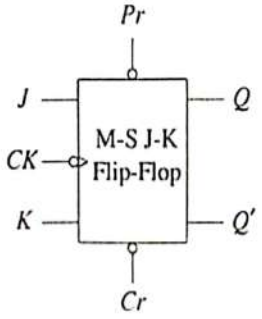
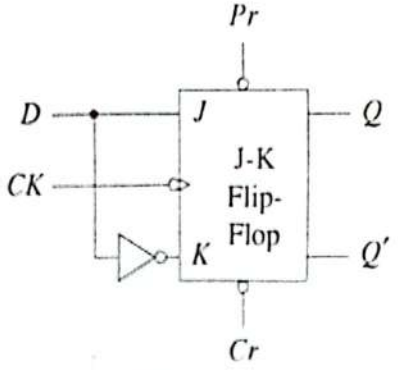


Figure 9.15 Logic symbol of M-S J-K flip-flop.

9.5 D Flip-Flop

If we convert a J-K flip-flop circuit as shown in Figure 9.16(a), we obtain a D flip-flop. The truth table of the circuit is shown in Figure 9.16(b).



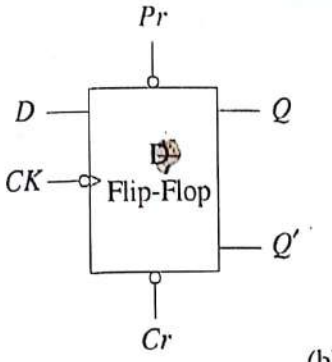
(a) Circuit connection

D_n	Q_n	J_n	K_n	Q_{n+1}
0	0	0	1	0
0	1	0	1	0
1	0	1	0	1
1	1	1	0	1

(b) Truth table

Figure 9.16 A J-K flip-flop converted into a D flip-flop.

The logic symbol of a D flip-flop is shown in Figure 9.17(a) and its characteristic table is shown in Figure 9.17(b). A reduced characteristic table is also shown in Figure 9.17(c) for convenience.



(a) Logic symbol

D_n	Q_n	Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

(b) Characteristic table

D_n	Q_{n+1}
0	0
1	1

(c) Reduced characteristic table

Figure 9.17 Logic symbol and characteristic table of a D flip-flop.

A D flip-flop has only one input referred to as D-input or data-input. From the characteristic table of Figure 9.17(c), it is clear that the output Q_{n+1} at the end of the clock pulse equals the input D_n before the clock pulse. This is equivalent to saying that the input data appears at the output at the end of the clock pulse. Thus, the transfer of data from the input to the output is delayed and hence the name *delay (D) flip-flop*. The D flip-flop is either used as delay device or as a latch to store 1-bit of binary information.

The characteristic table of Figure 9.17(b) is shown in the K-map of Figure 9.18, which gives the following characteristic equation of a D flip-flop

$$Q_{n+1} = D_n$$

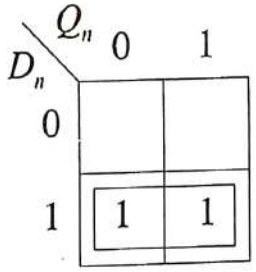


Figure 9.18 Characteristic equation of D flip-flop.

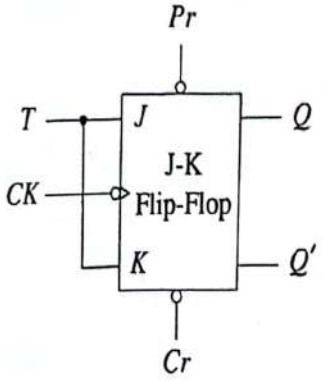
The excitation table of a D flip-flop can be easily derived from the characteristic table of Figure 9.17(b) and is shown in Table 9.6.

Table 9.6 Excitation table of D flip-flop.

Q_nQ_{n+1}	D_n
0 0	0
0 1	1
1 0	0
1 1	1

9.6 T Flip-Flop

If we convert a J-K flip-flop circuit as shown in Figure 9.19(a), we obtain a T flip-flop. The truth table of the circuit is shown in Figure 9.19(b).



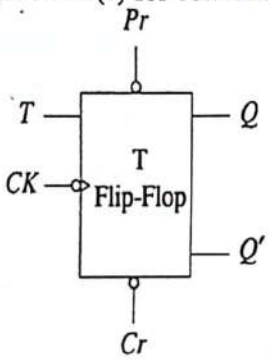
(a) Circuit connection

T_n	Q_n	J_n	K_n	Q_{n+1}
0	0	0	0	0
0	1	0	0	1
1	0	1	1	1
1	1	1	1	0

(b) Truth table

Figure 9.19 A J-K flip-flop converted into a T flip-flop.

The logic symbol of a T flip-flop is shown in Figure 9.20(a) and its characteristic table is shown in Figure 9.20(b). A reduced characteristic table is also shown in Figure 9.20(c) for convenience.



(a) Logic symbol

T_n	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

(b) Characteristic table

T_n	Q_{n+1}
0	Q_n
1	Q_n'

(c) Reduced characteristic table

Figure 9.20 Logic symbol and characteristic table of a T flip-flop.

A T flip-flop has only one input, referred to as T-input. From the characteristic table of Figure 9.20(c), it is clear that if $T = 1$ the flip-flop acts as a toggle switch. For every clock pulse, the output Q changes or toggles and hence the name *toggle (T) flip-flop*.

The characteristic table of Figure 9.20(b) is shown in the K-map of Figure 9.21, which gives the following characteristic equation of a T flip-flop

$$Q_{n+1} = T_n'Q_n + T_nQ_n'$$

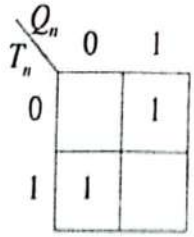


Figure 9.21 Characteristic equation of a T flip-flop.

The excitation table of a T flip-flop can easily be derived from the characteristic table of Figure 9.20(b) and is shown in Table 9.7.

Table 9.7 Excitation table of T-flip-flop.

$Q_n Q_{n+1}$	T_n
0 0	0
0 1	1
1 0	1
1 1	0

9.7 Conversion From One Type of Flip-Flop to Another Type

In earlier sections, we have discussed conversion from S-R to J-K, J-K to D, and J-K to T flip-flops. Now, we shall discuss the conversion from one type of flip-flop to another type by using a formal technique.

Consider the general model for conversion from one type of flip-flop to another as shown in Figure 9.22. In this, we are required to design the combinational logic for converting new input definition into input requirements for the given flip-flop that will cause the given flip-flop to perform as desired.

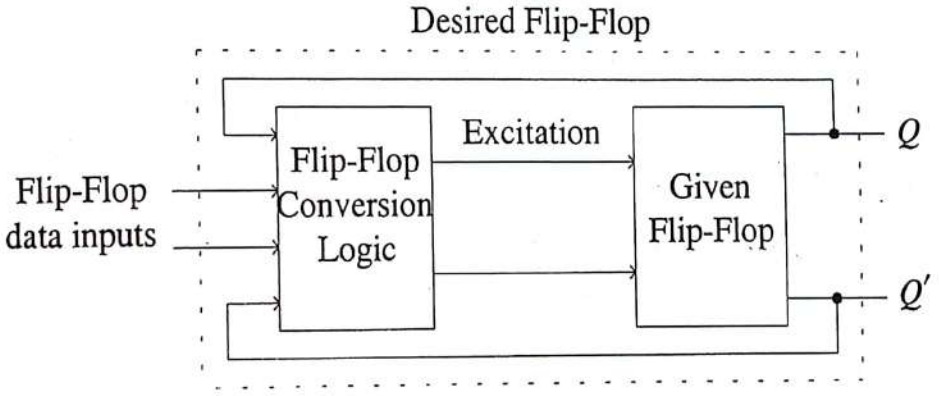


Figure 9.22 The general model used to convert one type of flip-flop to another type.

To design the conversion logic we need to make a truth table with data input(s) and Q as the inputs and the input(s) of the given flip-flop as the output(s). This is done by writing the characteristic table of the desired flip-flop and then determining the excitation requirements for the given flip-flop from excitation table of the given flip-flop. The conventional method of combinational logic design then follows as usual. The conversion of an S-R flip-flop into a J-K flip-flop is illustrated below.

The characteristic table of the desired J-K flip-flop and the excitation table of the given S-R flip-flop are combined in Table 9.8 to form the truth table for converting S-R flip-flop into a J-K flip-flop. The K-maps for S_n and R_n are given in Figure 9.23, which give

$$S_n = J_n Q_n'$$

$$R_n = K_n Q_n$$

Table 9.8 Truth table for converting an S-R flip-flop into a J-K flip-flop.

Characteristic table of the desired J-K flip-flop		Excitation required for the given S-R flip-flop
$J_n\ K_n\ Q_n$	Q_{n+1}	$S_n\ R_n$
0 0 0	0	0 x
0 0 1	1	x 0
0 1 0	0	0 x
0 1 1	0	0 1
1 0 0	1	1 0
1 0 1	1	x 0
1 1 0	1	1 0
1 1 1	0	0 1

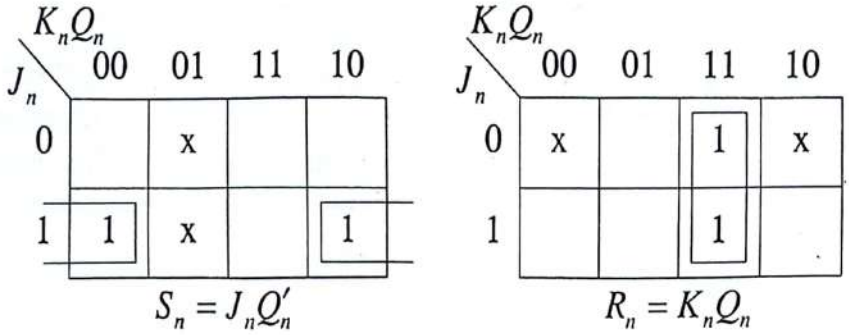


Figure 9.23 K-map for excitation function of S-R flip-flop for converting it into a J-K flip-flop.

Thus, we see that the circuit resulting from this design is the same as that shown in Figure 9.9(a).

9.8 Triggering of Flip-Flops

All flip-flops other than the master-slave type discussed earlier are level triggered, i.e., the outputs respond to the inputs (according to the characteristic table) as long as the clock is present.

The master-slave flip-flops are also referred to as the *pulse-triggered* flip-flop, i.e., the outputs respond to the inputs when a pulse is applied at the clock input. The master responds when the clock is present ($CK = 1$) and the output of the slave will be available at the falling edge of the clock pulse ($CK = 0$). As discussed in Section 9.4, this eliminates the problem of race-around condition. In this the data is *locked-out* at the falling edge of the clock pulse, i.e., the changes occurring at the inputs once *CK* goes to 0 will not affect the operation of the flip-flop.

The inputs to the flip-flop may change during the presence of the clock pulse due to certain operations in the system. This causes uncertainty in the outputs of the flip-flop, which is eliminated by using edge-triggered flip-flops. In the case of an edge-triggered flip-flop, the transfer of information from data input(s) to the output of the flip-flop occurs at the positive (or negative) edge of the clock pulse. The only

time the outputs can change state is during the brief interval of time when the clock signal is making a transition from the 0 to 1, or from the 1 to 0 states. A flip-flop that responds only to rising (or falling) edge is referred to as *positive-edge-triggered* (or *negative-edge-triggered*) flip-flop. The data lockout occurs at the end of the edge. The logic symbol used for an edge-triggered flip-flop is the same as that of a master-slave flip-flop.

The block diagram of an edge-triggered flip-flop is shown in Figure 9.24. The edge-detector produces a narrow positive pulse (CK^*) that occurs coincident with the transition of the *CK* input pulse. The output of the flip-flop responds to the inputs during this narrow pulse.

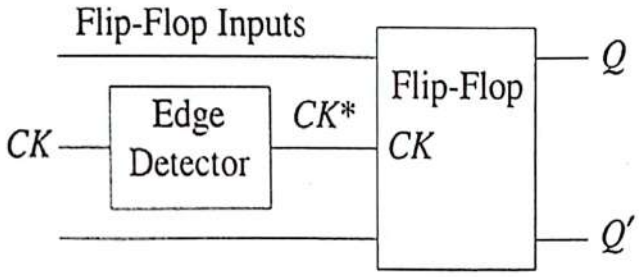
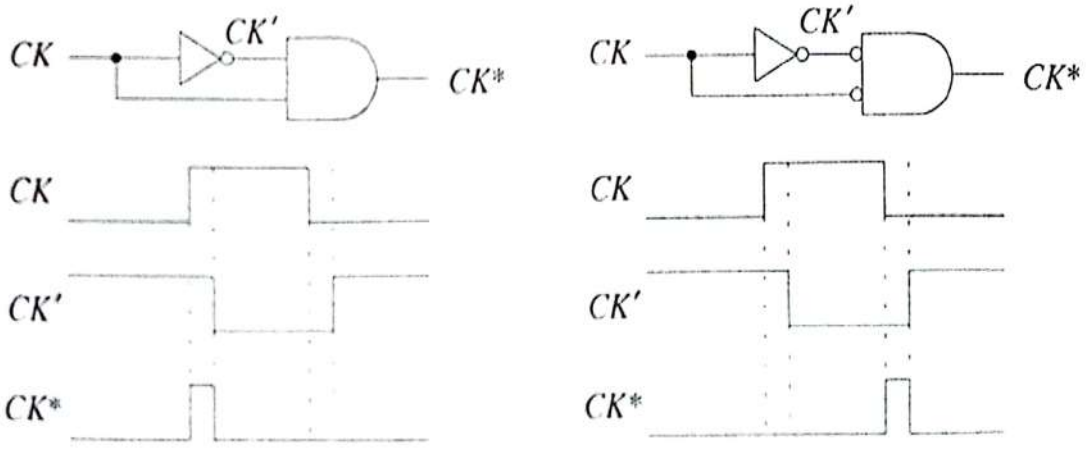


Figure 9.24 Block diagram of an edge-triggered flip-flop.

The generation of *CK** pulse for positive-edge-triggered flip-flop is shown in Figure 9.25(a). The inverter produces a delay of a few nanoseconds so that the transition of CK' occurs a little bit after that of *CK*. The AND gate produces an output pulse that is HIGH only for the few nanoseconds when *CK* and CK' are both HIGH. The result is a narrow pulse at *CK**, which occurs on the positive edge of *CK*. The generation of *CK** pulse for negative-edge-triggered flip-flop is shown in Figure 9.25(b). In this case, the AND gate produces an output pulse that is HIGH only for the few nanoseconds when *CK* and CK' are both LOW. The result is a narrow pulse at *CK**, which occurs on the negative edge of *CK*.



(a) Positive-edge detector (b) Negative-edge detector

Figure 9.25 Edge detector circuit.

Since the *CK** pulse is HIGH for only a few nanoseconds, the output Q is affected by the flip-flop inputs for a short time during and after the occurrence of the active-edge of the clock. This is what gives the flip-flop its edge-triggered property.

9.9 Available Flip-Flop ICs

Some available flip-flop ICs are listed in Table 9.9. Pin assignment of some of the useful Flip-flops is shown in Figure 9.26.

Table 9.9 Available flip-flop ICs.

IC No.	Description
7473	Dual J-K master-slave flip-flop
7474	Dual +edge-triggered D flip-flop
7475	Quad +level-triggered D flip-flop
7476	Dual J-K master-slave flip-flop
74107	Dual J-K master-slave flip-flop
74109	Dual +edge-triggered J-K′ flip-flop
74112	Dual –edge-triggered J-K flip-flop
74113	Dual –edge-triggered J-K flip-flop
74173	Quad +edge-triggered D flip-flop with 3-state output
74174	Hex +edge-triggered D flip-flop
74175	Quad +edge-triggered D flip-flop

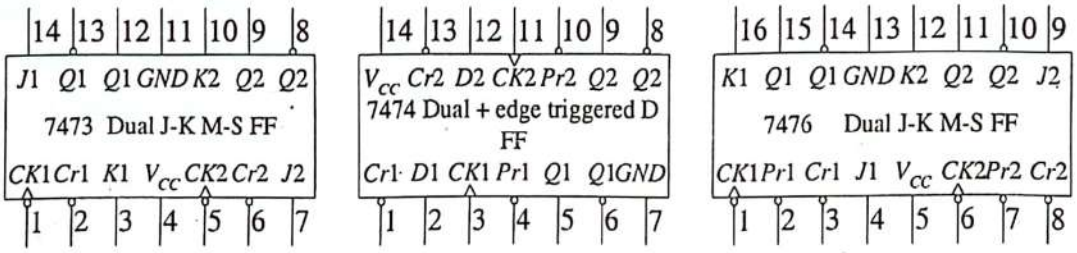


Figure 9.26 Pin assignment of some flip-flops.

9.10 Applications of Flip-Flops

The main application of flip-flops is as memory unit of sequential circuits. This application will be obvious in subsequent chapters. Here three more applications of flip-flops are discussed.

9.10.1 Bounce-Elimination Switch

Mechanical switches are employed in digital systems as input devices by which digital information (0 or 1) is entered into the system. There is a very serious problem associated with these switches, viz., *switch-bouncing* (or *chattering*). When the arm of the switch is thrown from one position to another, it chatters or bounces several times before finally coming to rest in the position of contact. The bounce is

the result of the spring-loaded impact of the switch throw contact and the pole contacts.

In a sequential circuit, if a 1 is to be entered through a switch, then the switch is thrown to the corresponding position. As soon as it is thrown to this position, the output is 1 but the output oscillates between 0 and 1 for some time due to make and break (bouncing) of the switch at the point of contact before coming to rest. This changes the output of the sequential circuit and creates difficulties in the operation of the system. This problem is eliminated by using bounce-elimination switches.

The circuit of Figure 9.27(a) acts as a bounce-elimination (chatterless) switch. The wave forms at S', R', Q, and Q' are illustrated in Figure 9.27(b). The switch (SW) is thrown from position A to B at $t = 0$. Therefore, at $t = 0^+$ the voltage at S' (A) will be V_{CC} (logic value 1) and will continue to remain so as long as the switch is not thrown to position A again. At R' (B), the voltage at $t = 0^-$ is V_{CC} (logic value 1) and goes to 0V (logic value 0) at t_1 (t_1 being the time delay of the switch). The switch arm makes contact at B at $t = t_1$, and then bounces off. Therefore, the level at R' changes from 0 to 1 and vice-versa. This is illustrated in Figure 9.27(b). Between $t = 0$ and $t = t_1$, both the inputs S' and R' are at 1 and therefore, Q does not change. The output Q changes at t_1 and becomes 0. Now, even when R' is changing at t_2, t_3, etc. Q does not change. This shows that it is a chatterless switch.

The latch used in Figure 9.27(a) can be replaced by the IC 74279, which is a quad S′-R′ latch.

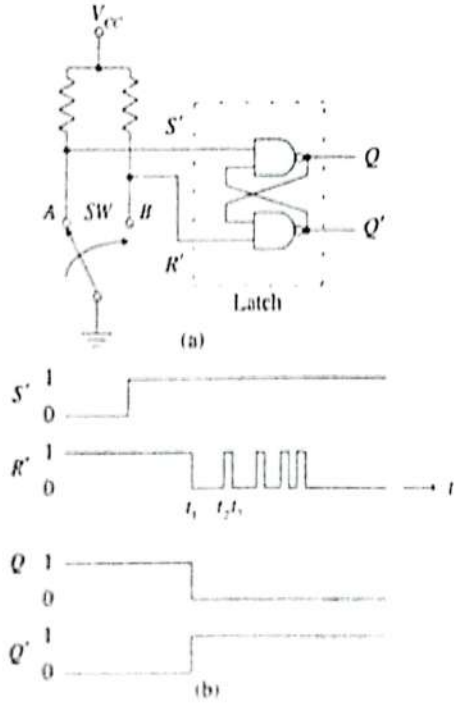


Figure 9.27 (a) A bounce-elimination switch, (b) waveforms of S', R', Q, and Q'.

9.10.2 Random-Access Memory

In computers, digital control systems, information processing systems, etc. it is necessary to *store* digital data and *retrieve* the data as desired. Flip-flops can be used for making memories in which data can be stored for any desired length of time and then read out whenever required. In such a memory, data can be put into (*writing* into memory) or retrieved from (*reading* from the memory) the memory in a random fashion and is known as *random-access memory*.

A 1-bit read/write memory is shown in Figure 9.28, which is the basic memory element and memory ICs are built around a system of basic 1-bit cell. In this memory cell, a level triggered D flip-flop is used which has Q output that follows the D input as long as CK terminal is at 1. The moment the CK input changes to 0, the Q output does not change and it retains the D input level that existed just before the transition from 1 to 0 at input CK. This input is used to select the memory cell. In the 1-bit cell shown there are three inputs – D_i (data input), A_n (address select) and R/W' (read/write control) and one output D_o (data output). $A_n = 1$ enables the cell for reading or writing operation, R/W' at 1 is for reading from the cell and 0 for writing into the cell. As long as $A_n = 0$, all input and output activities are blocked, and the cell is in the hold mode where its stored output is protected.

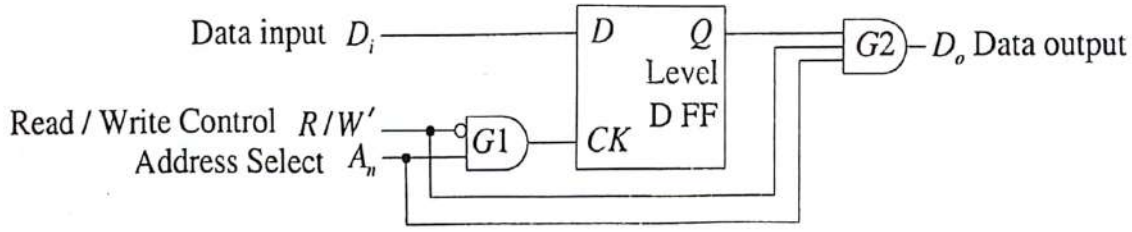


Figure 9.28 A 1-bit read/write memory cell.

The complete function of this cell can be understood from the function table of Table 9.10. The read operation is nondestructive, that is, the stored bit can be read out any number of times without disturbing it. The stored bit will be protected as long as power is on. Therefore, this type of memory is known as *volatile memory*.

Table 9.10 Function table of 1-bit memory cell.

Inputs			Mode
A_n	R/W'	D_i	
0	x	x	Hold, $D_o = 0$.
1	0	0	Write 0 into memory, $D_o = 0$.
1	0	1	Write 1 into memory, $D_o = 0$.
1	1	x	Read, D_o = stored D_i bit.

As far as writing into the cell is concerned, it is not required to be cleared before entering the new bit. Whenever a new bit is entered the earlier one gets destroyed automatically.

9.10.3 Frequency Divider

It is some times needed to generate low-frequency clock signal from a high-frequency clock signal. This can be achieved by the frequency divider circuit shown in Figure 9.29(a). Each T flip-flop has its T input connected to 1, so that it will change states (toggles) whenever the CK signal goes from 1 to 0. The high-frequency clock signal is applied to the CK input of flip-flop $FF1$. Output Q_1 is connected to the CK input of flip-flop $FF2$, and output Q_2 is connected to the CK input of flip-flop $FF3$. The waveforms of Figure 9.29(b) shows how the flip-flops change states as the clock pulse is applied.

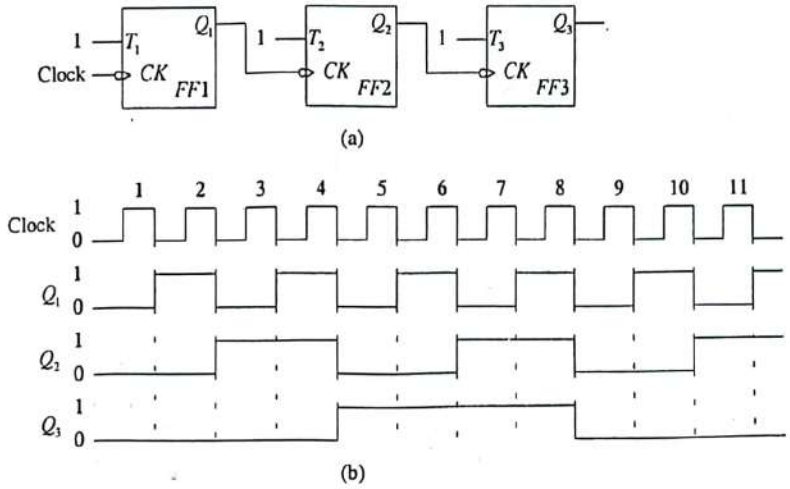


Figure 9.29 Frequency divider circuit using T flip-flops.

Flip-flop $FF1$ toggles on the negative-edge transition of each input clock pulse. Thus, the Q_1 output waveform has a frequency that is exactly one-half of the input clock frequency. Flip-flop $FF2$ toggles each time Q_1 output goes from 1 to 0. The Q_2 output waveform has a frequency equal to exactly one-half the frequency of the Q_1 output and, therefore, one-fourth of the input clock frequency. Flip-flop $FF3$ toggles each time Q_2 output goes from 1 to 0. Thus, the Q_3 output waveform has one-half the frequency of Q_2 output and, therefore, one-eighth of the input clock frequency.

As described above, each flip-flop divides the frequency of its input by 2. Thus, if we were to add a fourth flip-flop to the chain, it would have a frequency equal to one-sixteenth of the input clock frequency, and so on. Using the appropriate number of flip-flops, this circuit could divide a frequency by any power of 2. More specifically, using n flip-flops would produce an output frequency from the last flip-flop that is equal to $1/2^n$ of the input frequency.

Exercises

9.1 Draw circuits for M-S J-K flip-flop with NOR S-R flip-flop.

9.2 For the circuit shown in Figure 9.30(a) the clock and input waveforms shown in Figure 9.30(b) are applied. Sketch the waveforms of Q and Q' if the flip-flop is edge-triggered.

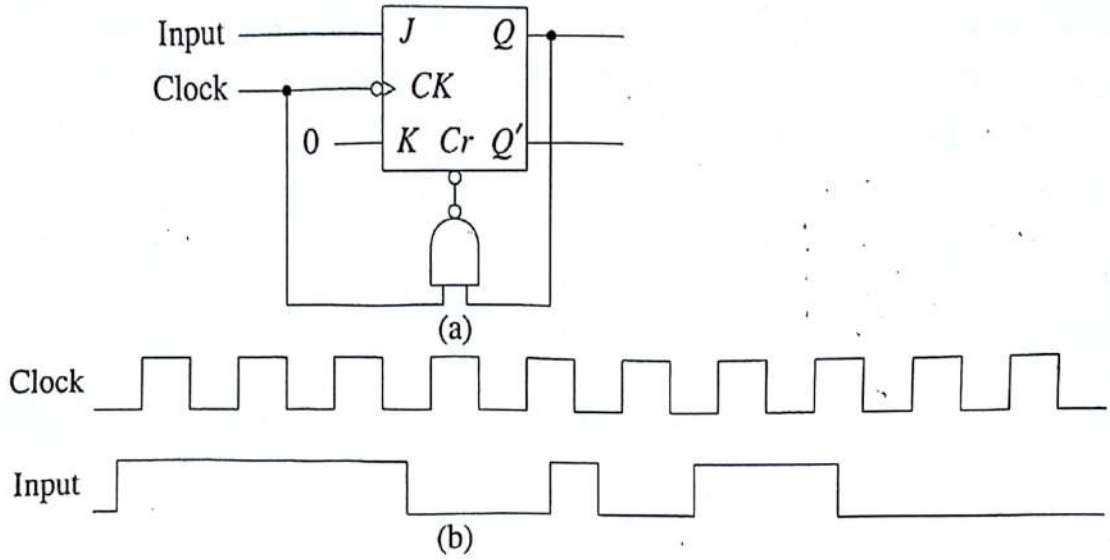


Figure 9.30 Figure for Exercise 9.2.

9.3 Repeat Exercise 9.2 if the flip-flop is master-slave.

9.4 If the waveforms shown in Figure 9.31 are applied to the circuit of Figure 9.30(a), sketch the output (Q) waveform assuming M-S flip-flop.

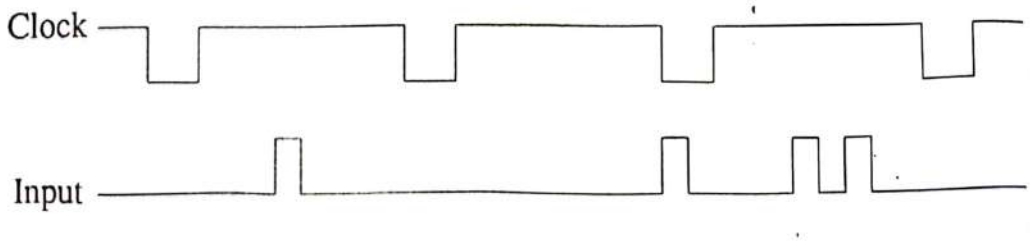


Figure 92.31 Figure for Exercise 9.4.

9.5 Verify that if Q' output of a D flip-flop is connected to D input, it acts as a toggle switch.

9.6 Perform the following flip-flop conversions:

(a) S-R flip-flop into D flip-flop.
(b) S-R flip-flop into T flip-flop.
(c) T flip-flop into D flip-flop.
(d) D flip-flop into T flip-flop.
(e) D flip-flop into J-K flip-flop.
(f) T flip-flop into J-K flip-flop.

9.7 Verify the operation of the debounce switches shown in Figure 9.32.

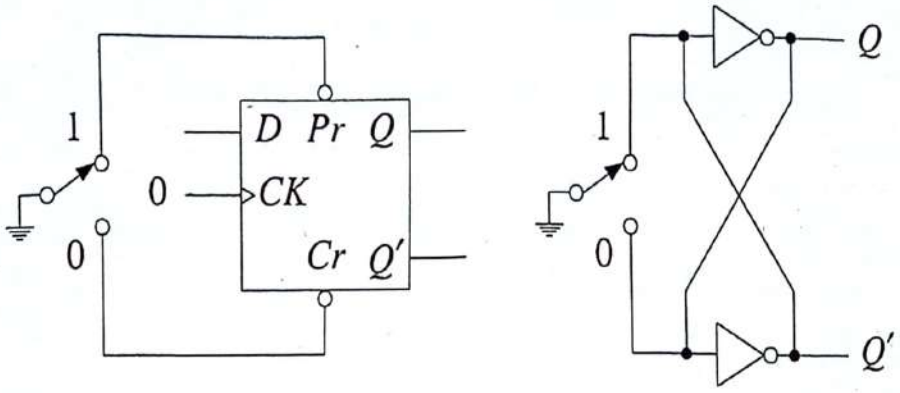


Figure 9.32 Debounce switches for Exercise 9.7.

Further Reading

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